



Assignment on recursion.

For each of the following problems, write the algorithm (pseudocode) of the function to solve it using the recursion indicating the **base case** and the **recursive step**. Implement these algorithms in C++.

NB: Algorithms must be written on the paper.

1. Sum of elements in an array.
2. Factorial of a number.
3. Compute the nth Fibonacci number.
4. Sum of digits of a number.
5. Find maximum element in an array.
6. Find minimum element in an array.
7. Reverse a string with recursion.
8. Check if a string is a palindrome.
9. Compute power of a number (x^n).
10. Count occurrences of a given element in an array.
11. Generate all permutations of a string.
12. Recursive linear search in an array.
13. Recursive binary search in a sorted array.
14. Check if an array is sorted recursively.
15. Print numbers from n down to 1
16. Calculate Greatest Common Divisor (GCD) of two numbers.
17. Print all subsets (power set) of a given set.
18. Recursive multiplication without using '*' operator.
19. Solve Tower of Hanoi puzzle.
20. Sum of even and odd numbers in a range.